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AGR: Reinforced Causal Agent-Guided Self-explaining Rationalization

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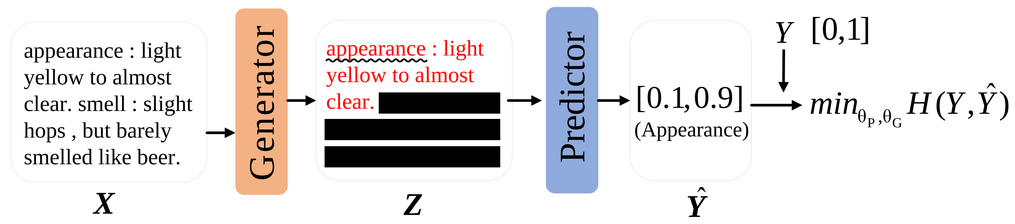
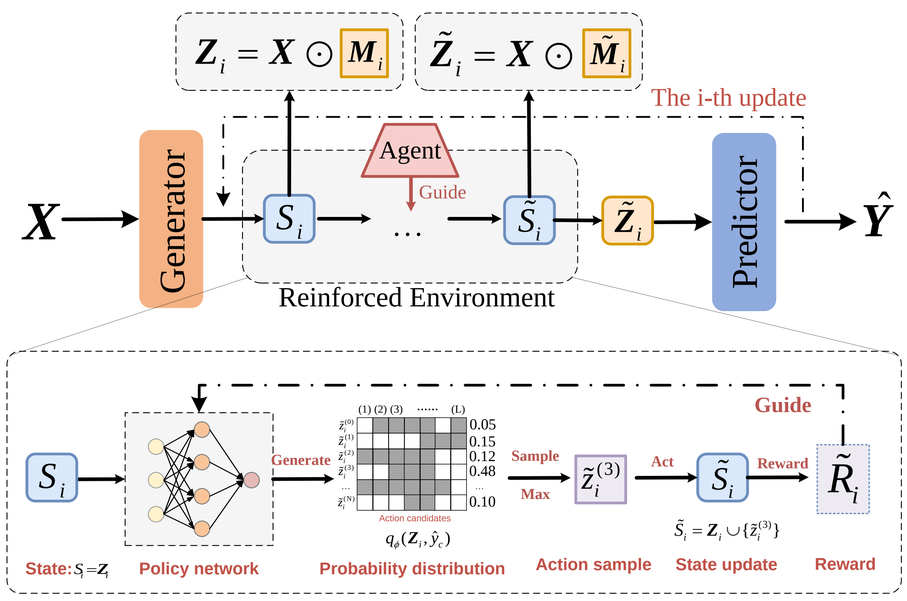
Cooperative Self-rationalization

As an interpretable paradigm, rationalization aims to *select a small and human-intelligible subset (i.e., rationale)* from the input to support and explain the prediction results when yielding them.



To alleviate degeneration problem, we propose **AGR** which leverages a reinforced causal agent to guide the cooperative game optimization.
1) We quantify the causal effects in the rationale optimization process, and design a reinforcement learning (RL) process (e.g., markov decision) to refine the learning bias during training.
2) We further pretrain an agent within reinforced causal environment to guide next actions by a system of rewards. We also theoretically illustrate that a robust model needs the desired guidance.
3) Experimental results demonstrate the effectiveness of our approach, surpassing state-of-the-art methods.

Methodology



The architecture of AGR.

Reinforced Causal Agent

1) Rationale Causal Attribution. Formally, we construct a rationale $\mathcal{Z}_k^*$ by maximizing an attribution metric $A(\cdot)$ in rationalization

$$\mathcal{Z}_K^* = \arg max_{\mathcal{Z}_K \subseteq X} A(\mathcal{Z}_K|\hat{y}_c), \tag{1}$$

where $A(\cdot)$ measures the contribution of each candidate $\mathcal{Z}_K$ to the target prediction $\hat{y}_c$. However, $A(\mathcal{Z}_K|\hat{y}_c)$ needs to be quantified. To this end, we introduce causal intervention calculus $do(\cdot)$, including $do(Z = Z_K)$ and $do(Z = \emptyset)$(Pearl, 2009; Pearl et al., 2016), and reformulate the causal contribution from $\emptyset$ to $\mathcal{Z}_K$ by mutual information, and quantify the learning objective between $\mathcal{Z}_i$ and $\mathcal{Z}_{i+1}$

$$A(\mathcal{Z}_K|\hat{y}_c) = I(\hat{y}_c, do(\mathcal{Z}_K)) - I(\hat{y}_c, do(\emptyset)). \tag{2}$$

$$\mathcal{Z}_{i+1}^* = \arg max_{\mathcal{Z}_{i+1} \in \{X \setminus \mathcal{Z}_i\}} A(\mathcal{Z}_{i+1}|\mathcal{Z}_i, \hat{y}_c). \tag{3}$$

2) Markov Decision Process as RL. Based on the markov decision process $(s_0, a_1, r_1, s_1, ..., a_K, r_K, s_K)$, we reformulate the sequence process as $(\langle s_0, \tilde{a}_0, \tilde{r}_0, \tilde{s}_0\rangle, a_1, r_1, \langle s_1, \tilde{a}_1, \tilde{r}_1, \tilde{s}_1\rangle, ..., a_K, r_K, \langle s_K, \tilde{a}_K, \tilde{r}_K, \tilde{s}_K\rangle)$, where $\langle s_i, \tilde{a}_i, \tilde{r}_i, \tilde{s}_i\rangle$ indicates process of transitioning from state s_i to $\tilde{s}_i$ in the *i-th* update. Given the state $s_i = Z_i$, we derive the available action space: $\tilde{\mathbb{A}}_i = \{X \setminus Z_i\}$. The searched action can be represented as

$$\tilde{a}_i = \tilde{z}_i, \tag{4}$$

where $\tilde{z}_i \in \{X \setminus Z_i\}$ indicates candidate rationale in action space. Having made the action $\tilde{a}_i$, the state transition is to merge $\tilde{z}_i$ into Z_i, i.e., $\tilde{Z}_i = Z_i \cup \{\tilde{z}_i\}$.

To assess the effectiveness of the action $\tilde{a}_i$ in mitigating the learning bias of the model, the reward $\tilde{\mathbb{R}}_i(\tilde{s}_i, \tilde{a}_i)$ at state s_i can be formulated as follows:

$$\tilde{\mathbb{R}}_i = \begin{cases} A(\tilde{z}_i|Z_i, \hat{y}_c^*) + 1, & if f_P(Z_i \cup \{\tilde{z}_i\}) = \hat{y}_c^* \\ A(\tilde{z}_i|Z_i, \hat{y}_c^*) - 1, & otherwise. \end{cases} \tag{5}$$

3) Pretrained Agent. We pretrain the auxiliary *agent*, denoted as $f_A(\cdot)$, with

$$\theta_A^* = arg \min_{\theta_A} \mathcal{H}(Y, \hat{Y}|X), s.t. \hat{Y} = f_A(X), \tag{6}$$

where θ_A indicates the *agent*, and θ_A^* denotes the optimal solution.

Lemma 1 *Given $X, Z, Y, \hat{Y} = f_P(f_G(X))$. Existing a guiding variable $\mathcal{A}$ could enable the predictor $f_P(\cdot)$ to achieve good predictions. That is, a solution for $\mathcal{A}$ exists, and X is a solution of $\mathcal{A}$. (For the process of proof, please see this paper.)*

Agent-Guided Rationalization

1) Policy Network Architecture. It takes the pair of intermediate state $\mathcal{Z}_i$ and $\hat{y}_c$ provided by $f_A(\cdot)$ as input. Formally,

$$\tilde{z}_i \sim q_\phi(\mathcal{Z}_i, \hat{y}_c), \tag{7}$$

Representation learning of action candidates. With the space of action candidates, our policy network first learns the representation for each action candidate $\tilde{a}_i^{(j)} (0 < j < N)$, where N is the number of candidates. Then, we employ the encoder to encode $X \setminus \mathcal{Z}_i$ for obtaining the action representation of $\tilde{z}_i$ by

$$e_{\tilde{z}_i} = encoder(X \setminus \mathcal{Z}_i), \tag{8}$$

Selection of action. The policy network aims to select a singular action $\tilde{a}_i = \tilde{z}_i$, prioritizing its relevance to the current state $s_i = \mathcal{Z}_i$. This selection process:

$$p_{\tilde{z}_i} = MLP([e_{\tilde{z}_i}; e_{\mathcal{Z}_i}]), \tag{9}$$

where $e_{\mathcal{Z}_i}$ indicates the current rationale's representation. The selection probability for each action candidate within $\tilde{\mathbb{A}}_i$ is computed using

$$\mathbb{P}_\phi(\tilde{z}_i|\mathcal{Z}_i, \hat{y}_c) = softmax_{\tilde{\mathbb{A}}_i}(p_{\tilde{z}_i}), \tag{10}$$

2) Policy Gradient Training. Since discrete sampling within the policy network blocks gradients, we adopt policy gradient-based training framework REINFORCE (Sutton et al., 1999). The objective $\max_\Omega(\mathbb{L})$ is as follows:

$$\max_\phi \mathbb{E}_{\mathcal{Z}_i \in \tilde{A}_i} \mathbb{E}_i[\tilde{\mathbb{R}}(\mathcal{Z}_i, \tilde{z}_i) log \mathcal{P}_\phi(\tilde{z}_i|\mathcal{Z}_i, \hat{y}_c)]. \tag{11}$$

The final task loss is a jointly optimized objective:

$$\min_{\theta_g, \theta_p} \mathcal{H}(Y, \hat{Y}) + \Omega(M) - \Omega(\mathbb{L}), s.t. \hat{Y} = f_P(f_G(X)) \tag{12}$$

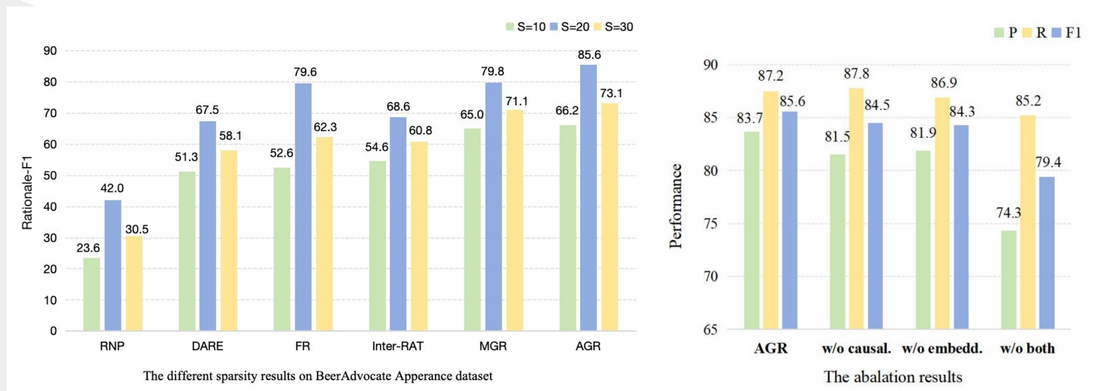
Experiments & Analysis

- Results on BeerAdvocate.

Methods	S	Appearance			Aroma			Palate		
		P	R	F1	P	R	F1	P	R	F1
RNP (Lei et al., 2016)	20	39.4	44.9	42.0	37.5	51.9	43.5	21.6	38.9	27.8
HardKuma (Bastings et al., 2019)	20	64.9	69.2	67.0	37.0	55.8	44.5	14.6	22.3	17.7
IB (Paranjape et al., 2020)	20	59.3	69.0	63.8	38.6	55.5	45.6	21.6	48.5	29.9
INVRAT (Chang et al., 2020)	20	58.9	67.2	62.8	29.3	52.1	37.5	24.0	55.2	33.5
DARE (Yue et al., 2022)	20	63.7	71.8	67.5	41.0	61.5	49.3	24.4	54.9	33.8
FR (Liu et al., 2022)	20	74.9	84.9	79.6	58.7	73.3	65.2	36.6	59.4	45.3
Inter-RAT (Yue et al., 2023)	20	62.0	76.7	68.6	44.2	65.4	52.8	26.3	59.1	36.4
MGR (Liu et al., 2023)	20	76.3	83.6	79.8	64.4	81.3	71.9	47.1	73.1	57.3
AGR(Ours)	20	**83.7**	**87.5**	**85.6**	**67.5**	**81.4**	**73.8**	**47.6**	**77.7**	**59.0**

- Results on HotelReview.

Methods	Location				Service				Cleanliness			
	S	P	R	F1	S	P	R	F1	S	P	R	F1
RNP (Lei et al., 2016)	10.9	43.3	55.5	48.6	11.0	40.0	38.2	39.1	10.6	30.5	36.0	33.0
CAR (Chang et al., 2019)	10.6	46.6	58.1	51.7	11.7	40.7	41.4	41.1	9.9	32.3	35.7	33.9
DMR (Huang et al., 2021)	10.7	47.5	60.1	53.1	11.6	43.0	43.6	43.3	10.3	31.4	36.4	33.7
A2R (Yu et al., 2021)	8.5	43.1	43.2	43.1	11.4	37.3	37.2	37.2	8.9	33.2	33.3	33.3
MGR (Liu et al., 2023)	9.7	52.5	60.5	56.2	11.8	45.0	46.4	45.7	10.5	37.6	44.5	40.7
AGR(Ours)	9.3	**54.9**	**60.5**	**57.6**	12.3	**45.9**	**49.3**	**47.6**	10.3	**39.0**	**45.5**	**42.0**



The different sparsity results and ablation results.